

Ziyan Lei

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SUMMARY

I am a motivated robotics student with hands-on experience in aerial robotics, UAV competitions, embedded control, and large-model research. My background combines practical engineering work with research exposure in areas such as model unlearning, bias evaluation, and visual perception. Through team projects and research internships, I have developed a careful and systematic working style, and I continue to refine my skills with the hope of contributing to high-quality research in the future.

RESEARCH EXPERIENCE

AISEC Research Group, Xi'an Jiaotong University

Sept 2023 – June 2025

- Worked under Prof. Chao Shen on large-model security, unlearning, and code-generation bias evaluation.
- Reproduced major LLMs (Stable Diffusion 2.1, Yi-34B, Qwen1.5-32B); experienced in distributed training and remote server workflows.
- Built an automated test pipeline and datasets for analyzing language bias in large code-generation models.
- Developed a knowledge-graph-based unlearning evaluation system; contributed to a paper accepted at AAAI 2026.

PROJECTS

Challenge Cup – eVTOL Exploration (National Grand Prize)

Jul 2024 – Dec 2024

- Vision subsystem lead: trained YOLOv8s model achieving 100% detection accuracy with > 0.82 confidence.
- Implemented maritime target coordinate solving and route optimization for autonomous rescue missions.

RoboCup China – Drone Challenge (National First Prize)

Jan 2024 – May 2024

- Built onboard recognition and classification using OpenMV + YOLO.
- Developed real-time obstacle avoidance and visual-assisted navigation algorithms for robust flight.

SSVEP Brain-Computer Interface System (Team Leader)

May 2024 – Jul 2025

- Designed an experimental paradigm that leverages large-model-guided optimization to enhance EEG/SSVEP signal analysis, enabling more reliable feature extraction and interpretation.
- Improved SSVEP classification and typing efficiency; produced a patent (expected approval within three years).

RoboMaster Robotics Competition – Infantry Robot Development

Sept 2022 – June 2025

- Developed motion control and basic vision-based targeting modules for the infantry robot using PID/LQR and vision feedback to ensure agile and stable operation in competition.
- Designed an STM32-based bidirectional BUCK-BOOST supercapacitor power module to improve power efficiency and system reliability (developed later in a technical advisory role).

PUBLICATIONS

1. From Chaos to Clarity: A Knowledge Graph-Driven Audit Dataset Generation Framework for LLM Unlearning

with Prof. Chao Shen and collaborators.

Accepted at AAAI 2026.

arXiv: [2502.18810](#)

2. Exploring Language Bias in Large Code Generation Models

with Prof. Chao Shen and collaborators.

arXiv: [2406.00602](#)

EDUCATION

2025 – 2026 MSc in Aerial Robotics, **University of Bristol**

2021 – 2025 B.Eng. in Automation, **Xi'an Jiaotong University** (GPA: 87/100)

AWARDS

National Grand Prize – Challenge Cup

National First Prize – RoboCup China Drone Challenge

National Undergraduate Innovation Training Program (x2)

First Prize – National Mathematical Modeling Competition (Shaanxi Division)

H Award – American Mathematical Modeling Contest

Cyrus Tang Foundation Scholarship (4 years), Outstanding Student Award

SKILLS

Programming Python, C/C++, MATLAB, PyTorch, TensorFlow, Git, CMake

Embedded Systems STM32 development, sensor drivers, real-time control

Tools Linux, remote server workflows, CUDA training pipelines

Languages English: IELTS 6.5, CET-6 560+, GRE 327 (AW 4.0);

German: A2